20 The table shows some values of x and y that satisfy the equation $y = a\cos x° + b$

x	0	30	60	90	120	150	180
y	3	$1 + \sqrt{3}$	2	1	0	$1 - \sqrt{3}$	-1

Find the value of y when $x = 45$

.....

(Total for Question 20 is 4 marks)

21 Show that $\dfrac{6 - \sqrt{8}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

√8 = 2√2. Multiply by the conjugate:

$$\frac{6 - 2\sqrt{2}}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

1 mark
Multiply top and bottom by √2 + 1

×	√2	+1
6	6√2	+6
−2√2	−4	−2√2

1 mark
Expand the numerator completely

Denominator: 2 − 1 = 1

Numerator: 6√2 + 6 − 4 − 2√2 = 2 + 4√2.

Dividing by 1 gives 2 + 4√2, so a = 2 and b = 4.

1 mark
Correct integer coefficients

(Total for Question 21 is 3 marks)

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